

Instructor: Date:

Learner: Regional Branch:

Program: Center:

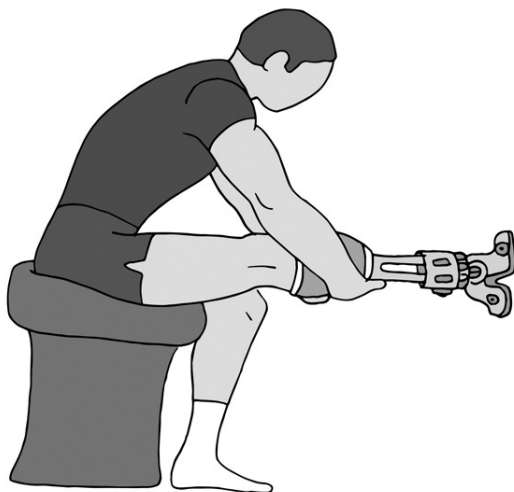
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Let's Start!

Prosthetics have come a long way!



Learning Outcome

Describe events in the past through a wider range of B1 language in both work and social contexts.

Ask your partner:

- What do you know about prosthetics?
- What do they look like today compared to what they were like in the past?

Can Do Statements

- I can talk about technology and subjects of study.
- I can organize points in an argument.
- I can express possibility and certainty.





Let's Explore!

Now you can be fitted with a bionic **prosthesis***!

* A "prosthesis" is an artificial device that replaces a missing part of the body.



Instructions: Read the text aloud.

Although it might sound like science fiction, the idea of replacing human body parts with machines is actually part of a science called bionics.

Bionics, in general, is the study of how the characteristics of living systems can be used to solve engineering problems. First, studies of how birds fly were used to improve airplane design. Then, scientists studied how the eye worked, so they could design better cameras.

Now, bionic scientists are studying how human body parts function. Of course they can use this information to design artificial parts to replace human ones.

In fact, bionics can be used very successfully in the area of artificial limbs. Clumsy prosthetic limbs are being replaced with artificial limbs that are electronically controlled. They work through electrical signals which are received from the muscles where the skin is in contact with the prosthesis.

Electrodes on the surface of the skin measure muscle movement and transfer that information to the bionic limb. Of course, these artificial limbs behave more like real ones. In fact, those who use one of these prostheses feel as if they have truly regained the use of a limb.

Of course, today, many body parts can be replaced by artificial ones. In fact, bionic scientists believe that one day, many more parts of the human body will be replaceable.





Let's Practice!

Let's talk about bionics!

Instructions: Read the article again. Select the incorrect sentences.

1. Which sentences talk about bionics?
 - a. If genetic engineering expands into gene transfer and recombination, it will create new viruses and bacteria.
 - b. Bionic scientists are studying how human body parts function.
 - c. Clumsy prosthetic limbs are being replaced with artificial limbs that are electronically controlled.
 - d. Even if we only meddle with nature in a minor way, it will certainly bring serious consequences.
 - e. Electrodes measure muscle movement and transfer that information to the bionic limb.



Let's Practice!

Lana is organizing some ideas to write an essay.

Instructions: Organize the mystery sentences. Read the sentences aloud. Check your pronunciation with your instructor.

1. used / design. / fly / studies / airplane / how / First, / to / improve / birds / of / were
2. Then, / studied / better / so / they / cameras. / how / design / worked, / the / scientists / eye / could
3. Now, / body / are / parts / how / scientists / studying / function. / human / bionic
4. to / they / can / this / parts / ones. / use / to / human / artificial / replace / Of / design / information / course
5. very / limbs. / successfully / the / in / can / be / artificial / fact, / In / used / bionics / of / area



Extension Activity

Listen to a TED Talk on prosthetic limbs.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Go online and find a TED Talk on prosthetics.
- Listen to the TED Talk and take some notes.
- Highlight any technical language.
- You may need to listen more than once.
- If there are subtitles, you can read along in English.
- Write a paragraph saying what you have found out about prosthetics.
- Share what you have learned with a partner or with your class.

Self-Assessment

Yes No

I listen to a TED Talk on prosthetics.

☐ ☐

I make notes, highlighting technical language.

☐ ☐

I share my paragraph and discuss my findings.

☐ ☐


Target Words

- to replace** (verb): to take the place of something that was there before
e.g. Bionic scientists can use information to design artificial parts to replace human ones.
- bionics** (noun): the design of electronic devices and mechanical parts based on how humans and animals perform certain tasks and solve certain problems
e.g. Bionics can be used very successfully in the area of artificial limbs.
- characteristic** (noun): a distinguishing feature or quality
e.g. Bionics is the study of how the characteristics of living systems can be used to solve engineering problems.
- system** (noun): a set of connected things that work together for a particular purpose
e.g. The study of living systems helps scientists with the development of bionics.
- to function** (verb): to perform a specified action or activity
e.g. Bionic scientists are studying how human body parts function.
- clumsy** (adjective): awkward in movement or action; without skill or grace
e.g. Clumsy prosthetic limbs are being replaced.
- electronically** (adverb): controlled by a piece of electronic equipment, especially a computer
e.g. Clumsy prosthetic limbs are being replaced with artificial limbs that are electronically controlled.





Learning Points

Organizing Points in an Argument

Discourse markers show how ideas are connected. They usually come at the beginning of a sentence.

Discourse markers can show a **connection** to what the speaker is saying and what was just said. They can also show what the speaker's **opinion** is about what is being said.

Some discourse markers include:

Discourse Marker	Meaning
although	in spite of the fact that
first	in the beginning
then	after that
now	currently
of course	as everybody knows, as is obvious
in fact	I want to emphasize

Examples:

- *Although it might sound like science fiction, the idea of replacing human body parts with machines is actually part of a science called bionics.*
- *First, studies of how birds fly were used to improve airplane design.*
- *Then, scientists studied how the eye worked, so they could design better cameras.*
- *Now, bionic scientists are studying how human body parts function. Of course they can use this information to design artificial parts to replace human ones.*
- *Of course, today, many body parts can be replaced by artificial ones.*
- *In fact, bionic scientists believe that one day, many more parts of the human body will be replaceable.*



Learning Points

Modals for Possibility and Certainty

You can use modals to talk about degrees of certainty.

You use them to describe what is definite, probable, possible, or impossible from the speaker's point of view.

You can use *must*, *may*, *might*, and *could* to describe degrees of certainty.

Must = it's very likely

Must not/Could not/Can't = it's not possible

May/Might/Can/Could = it's possible

May not/Might not = it's possible this is not true

You use modal verbs to add meaning to other verbs. They come before the dictionary form (infinitive without *to*) of the main verb. Modal verbs can add meaning connected with certainty or obligation.

You can use modal verbs in the passive voice.

You use the passive voice when you want to focus on the "receiver" of the action, rather than the "doer" of the action.

Modal	Form
They can use this information.	active modal
They can be used to solve engineering problems.	passive modal

Examples:

- *Then, scientists studied how the eye worked, so they could design better cameras.*
- *Bionics can be used very successfully in the area of artificial limbs.*
- *Today, many body parts can be replaced by artificial ones.*
- *Bionic scientists believe that one day, many more parts of the human body will be replaceable.*

